Perpendicular Lines — Practice — Answer Key

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Right Angles and Perpendicular Lines

Watch Me — Worked Example

$\overleftrightarrow{AB} \perp \overleftrightarrow{CD}$ at E, and $m\angle AEF = 35°$ with $\overrightarrow{EF}$ between $\overrightarrow{EA}$ and $\overrightarrow{EC}$.
$\angle AEC = 90°$, so $m\angle FEC = 90 - 35 = 55°$. **Answer: 55°**

1. $a \perp b$. What is the measure of each of the four angles they form?

$90°$

2. Two lines form a linear pair of congruent angles. What can you conclude?

the lines are perpendicular

3. Two lines are perpendicular. One angle is $(3x + 15)°$. Find x.

$x = 25$

4. $\overrightarrow{BD}$ is inside right angle $\angle ABC$. $m\angle ABD = (2x)°$, $m\angle DBC = (x + 18)°$. Find x.

$x = 24$

Type 2 Perpendicular to the Same Line

Watch Me — Worked Example

In a plane, $p \perp t$ and $q \perp t$. Both make $90°$ corresponding angles with t, so $p \parallel q$.

5. In a plane, $a \perp c$ and $b \perp c$. How are a and b related?

$a \parallel b$

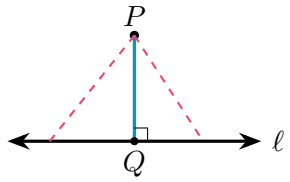
6. $a \parallel b$ and $c \perp a$. How are c and b related?

$c \perp b$

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Type 3 Distance From a Point to a Line

Watch Me — Worked Example



Point $(-2, 7)$ and the line $y = 3$. Straight down to $(-2, 3)$: distance $= 7 - 3 = 4$.
Point $(5, 1)$ and the line $x = -1$. Straight across to $(-1, 1)$: distance $= 5 - (-1) = 6$.

7. Distance from $(4, 9)$ to the line $y = 2$.

7

8. Distance from $(-3, -1)$ to the line $y = 4$.

5

9. Distance from $(6, 2)$ to the line $x = 1$.

5

10. Distance from $(-4, 0)$ to the y-axis.

4

11. From P, the segments to line ℓ have lengths 9, 6, 7.5 and 11. Which could be the distance from P to ℓ?

6 **(the shortest)**

12. HARD The parallel lines $y = 2x+1$ and $y = 2x+6$ are 5 apart measured straight up. Is the distance between them 5? Why?

no — a vertical segment is not perpendicular to them, so the distance is less than 5

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